World Simulator Checklist

Oct 14, 2026 · @Daniel

Where we are now

Phases 1 to 6 are finished and tested, Step 1 of the build order (peacetime politics) is done, and Step 2, the chronicle, is done: every big event the game can produce is now written down. Step 3, the foundations war needs, is done too. The next build is Step 4, starting with the kingdom rework (regions, countryside, lords) that real war will stand on.

Phase	Status	What is still open
1-2 Foundation and simulation core	Done	Seed Manager works informally, not formalized
3 World structure	Done	Nothing
4 NPC life	Done	Walking between buildings; traveling NPCs still earn wages
5 Economy	Done	Building upgrades (Step 4)
6 Relationships and memory	Done	Nothing
7 Events and history	In progress	Disasters, festivals, crime, monster raids
8 and 13 Kingdom agency and the chronicle	In progress	Expansion and captured capitals (Step 4), Eras (Step 5)
9-12, 14-15 Combat depth, Map, Decision Engine, AI agents, mods, polish	Not started	Armies and real war (Steps 3 and 4), Decision Engine and AI (Step 5), Map, mods, polish

Build order

All open scheduled work, in the order we agreed: peacetime politics, then the chronicle, then the foundations war needs, then real war, then the long-term systems.

Step 1: Peacetime politics

- [x] ~~Alliance dissolution: an alliance ends when either kingdom's favor falls below 40~~
- [x] ~~Succession shocks: a new monarch keeps half the kingdom's old feelings and re-rolls the rest (a quarter if the old ruler was overthrown)~~
- [x] ~~War chosen by grievance: a kingdom with a bitter enemy fights more often (4% to 12% a year) and favors the neighbor it resents~~
- [x] ~~War moves favor: the defender loses 30 toward the attacker (50 if a city falls), the attacker loses 10~~

Step 2: The chronicle

- [x] ~~chronicle_entries table: kingdom, event type, text, year, season (plus an empty era column and a link back to the raw event)~~
- [x] ~~Olde-English templates in Data/History/ChronicleTemplates.json: three wordings per event, chosen the same way every time for the same event~~
- [x] ~~Entries: Alliance Formed and Alliance Ended, written for both kingdoms, each from its own side (the kingdom that cooled withdraws its hand; the other is forsaken)~~
- [x] ~~Entries: ruler deaths by cause (old age, war naming the enemy, civil war, raid, starvation; reigning kings and queens only) and every coronation: each realm's founding monarch, successors, and those raised after an overthrow~~
- [x] ~~Entries: war from each side's point of view (win and lose, attacker and defender)~~
- [x] ~~Entries: civil wars, recorded by the kingdom itself and naming the sovereign the people rose against (captured capitals moved to Step 4)~~

Step 3: Foundations for war

- [x] ~~Consumption sinks: Tools wear out (output falls without them), Cloth is worn out by the population, buildings wear and are repaired with Wood and Stone, Potions heal the wounded and slowly spoil; cities start with stocked storehouses based on their population~~
- [x] ~~Army movement: an army is one record (roster, carried Food, Weapons and Potions, position on a road), not hundreds of individual travelers. It marches the quickest road route at road speed, eats from its wagons (an unfed army goes hungry), and disbands only at home; deployed soldiers are paid but not fed from the capital~~
- [x] ~~Supply lines: an army in the field asks its capital for supplies when its Food runs low (earlier the farther it is from home), and a supply caravan follows it along the roads and loads its wagons; a raid (3% a day) loses the cargo, so cutting a line is a real tactic~~

Step 4: Real war

- [x] ~~Regions layer: the map is cut into regions about 250 units across (64 today, more on a bigger map), each with terrain, climate by latitude, rivers flowing downhill, farmland quality and a name; every settlement sits in one. read_regions.py prints a text map. Nothing drawn yet; the visual map comes later~~
- [x] ~~Countryside A, villages and where they sit: settlements are placed where the land can feed them (no capital in the mountains); villages hold peasant households (5 people each) so about 70% of people are rural, bigger on better land, plus named villagers (about 4 x the square root of households, 8 to 60) as individual NPCs. Households are counted as people but neither eat from stock nor produce until harvests exist. Roughly halves the individual NPCs, and a simulated day takes about half as long~~
- [x] ~~Countryside B, harvests and grain: at the start of Autumn each village reaps households x 3,600 x farmland x weather (good 1.2, bad 0.55; bad about 1 year in 5 per region, shared with neighbors, sometimes two running, and about once in 16 years across most of the world). 20% kept as seed, 10% grain tax to the capital, the households keep a year's food in their own store, and the surplus goes to market, which trade carries to the towns. A normal year leaves a kingdom about 8% above need. Households eat 5 a day from their store, then buy from market; trade counts them as buyers once their store runs low, so poor-land villages are fed from elsewhere. Empty store and market counts hungry days (consequences come with eating from stock)~~
- [x] ~~Countryside C, entering the story: a villager becomes an individual NPC, and stays one. Famine flight: a village hungry 10 days running loses 5% of its households a month to the best-fed town with free houses, each arriving as a married couple with three children among the town's poor. Banditry: hungry 20 days running, a 50% chance a month that 1 to 3 young men join the nearest Bandit Camp. Levies: raise_army(levies=N) draws young men from the villages as Soldiers, who go home as Farmers when the army disbands. Each person who leaves takes a fifth of a household. Open question: towns are nearly full, so flight is capped by free houses; later, refugees could arrive homeless (vagrants) or building upgrades could make room~~
- [x] ~~Eating from stock, part 1 (done): each household (everyone sharing a home) buys its day's food at the city's price from the city's stock with its adults' pooled gold (children earn nothing). Jobless adults earn 3 gold a day as labourers. A household that cannot pay gets alms where the city has a working temple and charity funds (the city's wealth, filled by food sales). Meals restore only the share of hunger a household could afford, and going without costs stress and happiness, feeding the existing starvation and civil-war rules. New worlds: adults start with 30 days of food money, and markets~~

~~hold last year's surplus until the first harvest. In ordinary times everyone eats. Part 2 (done): each kingdom has a food price level on top of the up-to-2x scarcity price. A bad harvest raises it by 20% the first year in a row, 40% the second, 60% the third (about 2.7x after three); a good harvest eases it a quarter of the way back; ordinary years hold it; a revolt the rebels win resets it. Wages stay fixed, so years of bad harvests squeeze families (a one-earner family of five can buy 65% of its food at 2.7x, 33% if markets are thin too). After 20 hungry days running, village households die at up to 0.3% a day (about 9% a month), logged monthly~~

- [x] ~~Lords and provinces: crown, then hereditary great lords, then villages. Lords have their own loyalty, families and succession, send taxes and troops up, and can rebel or break away. Part 1 (done): every town besides the capital is a County held by a Count or Countess; villages fall under the nearest seat or, nearest the capital, the crown. On a lord's death the eldest grown child (16+) inherits and moves to the seat, else the surviving spouse, else the crown appoints a new lord who founds a new house; inheritances and appointments are logged. Part 2 (done): crowns and counties stay in the family. Line of succession: children (eldest first), the deceased's siblings, the spouse, the reigning monarch's siblings, the spouse's siblings, other close kin eldest to youngest, then the people (a new house). A child heir is crowned with a regent (their surviving parent, else the first adult in the line, else the realm's eldest Count or Countess) until 16; regencies and comings of age are chronicled. A lord who inherits the crown hands their county down their own line; a revolt starts a new dynasty. Lords keep 30% of their county's coin and grain taxes; the crown gets the rest. Each lord's loyalty drifts monthly toward a target: 55, +20 if kin of the monarch, -10 to +10 for their feeling toward the monarch, +5/-10 for a war won or lost in the last two years, up to -20 for famine in the county, -10 per step of bread prices above normal; below 30 a lord is restless (logged). Part 3 (done): each month a restless lord may rise (15% at 0 loyalty, falling to 0 at 30), and every other restless lord of the kingdom joins them; each rebel lord spends their purse on mercenaries (500 gold each, fighting as the crown's average soldier), and calls up levies (one per 10 households). The coalition rises only with at least a third of the crown's strength. Crushed: every rebel lord is executed and each county goes to a new house. Victorious: the counties break away together as a new kingdom under the leader (the leader's county becomes its crown lands, the allies' stay counties under their lords at 70 loyalty), starting at -30 favor with the old crown; chronicled from both sides. Lords spend a tenth of their purse each month, so savings settle near ten months' income (about 20 mercenaries)~~

- [x] ~~Kingdom health (Loyalty, Stability, Unrest) computed from what the sim already tracks, and the ruler's edict menu (taxes, festivals, granary, export bans, martial law) chosen by personality. Part 1 (done): every NPC has three traits from -100 to +100 (generosity: generous to greedy; mercy: merciful to harsh; boldness: bold to cautious), set once; children take after both parents (their average, varying by about 20). Each month every kingdom's health is worked out: Loyalty from its lords' loyalty weighted by county size; Unrest from stress, unfed food, bread prices and hungry villages; Stability from unrest, restless lords, regency, a recent rising or civil war, and raids. Stability below 30 is a crisis (logged once). Part 2 (done): the ruler (or regent) reviews the realm every season, and at once in a crisis, issuing at most one edict. Each edict's need comes from the realm (famine: granary, export ban; unrest: festival, lower taxes, martial law; empty treasury: raise taxes; collapsing stability: martial law; in calm times a ruler with a strong trait, about 50 or more, is tempted by their own nature), weighted by personality (generosity favors lower taxes, festivals and the granary; greed favors raising taxes; harshness and boldness favor martial law; caution favors the export ban). No repeat within a year except in a crisis. Effects: tax rate +-5% (2% to 30%), lords' loyalty +-5 per step, happiness +-5; festival +15 happiness, -15 stress for 1 gold and half a day's food a head; granary sends up to half the capital's Food to the hungry; export ban stops Food sales abroad for a season, -5 favor from every other kingdom; martial law for a season makes risings and civil war a quarter as likely, +10 stress, -10 happiness, 2 gold a soldier. Edicts are logged; tax changes, the granary and martial law are chronicled with the ruler's nature~~
- [x] ~~Kingdom types: agrarian crowns and merchant republics first; horse lords and tribute empires later. Part 1 (done): a quarter of a new world's kingdoms (rounded; at least one in a world of two or more) are merchant republics. A republic's Doge is elected for life by its Great Council (the capital's 20 wealthiest adults) from its 5 richest notables, votes swayed by wealth, each councillor's feelings and chance; never inherited, so no regency; elections are chronicled in their own wording (founding, succession, after an overthrow). Republics have no counties or lords (towns keep appointed Mayors), and half their people live in the countryside rather than 70%. The Doge issues edicts like any ruler. Part 2 (done): customs duties paid by foreigners -- a kingdom buying from a republic pays 10% on top, and a republic buying abroad keeps 5% of the price (the seller receives 95%); and a grain office that each month tops up the capital's reserve toward 60 days of food for the republic's townsfolk, buying from the foreign kingdoms with the biggest surplus (never one that has banned food exports), spending at most a quarter of the treasury a month, with the grain sent by caravan (it can be raided). Horse lords and tribute empires come later~~

- [x] ~~Campaigns that march to and fight at a real location, spending treasury gold, food, potions and tools while they are there. Soldiers away with an army must also stop counting toward their home city's defense in kingdom wars and civil wars (already fixed for raids). Done: a war declared in the yearly tick raises a campaign at the capital (two thirds of the free soldiers, a third staying home, plus levies of one per 20 households from the crown's villages and every loyal county; restless lords send none) and marches by road on the enemy's richest reachable town, eating from its wagons with supply caravans following. On arrival it fights the town's garrison plus every enemy soldier within 5 days' march (soldiers away with an army do not come), with the usual war aftermath; victory takes the town, and a county seat brings its whole county and lord (loyalty 40; a republic dissolves the county). The wounded drink the army's potions, then it marches home and disbands. Soldiers away with an army no longer defend home in wars or civil wars. Not yet spent while there: tools (sieges, next item)~~
- [x] ~~Armor, sieges and occupation (Phase 9). Part 1, armor (done): a new resource made at Armories (capitals and cities 1-2, towns 0-1) by Armorers, 2 Steel a suit; worlds start with 0.03 suits a person. Armies take up to a suit per soldier from the capital; defenders use their towns' stock. A side's armor share (suits per fighter, at most 1) raises its strength by up to half and cuts its deaths and injuries by up to 40%; the fallen's armor is lost. Raids stay unarmored. Balance note: Tools were already falling slowly in a test world before armor (-17 a day); armories add about -3 Tools and -17 Steel a day; worth a balance pass. Part 2 (done): capitals and towns with a Castle are walled (cities and towns now get 0-1 Castles; before, only capitals had one, so no siege could ever happen); defenders behind standing walls fight at double strength. An army reaching a walled town besieges it: the town is blockaded (no trade in or out), and siege works grow by a quarter point per soldier a day plus 20 per Tool spent (an army carries a Tool a soldier and spends 1 per 20 soldiers a day): about 7 days to breach 1000 points with Tools, 34 without. Breached: stormed without the walls. Ten days without Food: stormed, the starving garrison at half strength (cancelling the walls). After 120 days, or a week without food, the siege is lifted. Each step is logged. Still open: caravans already on the road can still deliver into a besieged town. Part 3 (done): a town taken by a campaign is occupied by a garrison of a quarter of the surviving army (soldiers before levies), who move in, stay soldiers and leave the army. Each month the townsfolk lose 5 happiness and gain 5 stress, and have a 15% chance to rise if they can muster at least half the garrison's strength (a tenth of the townsfolk plus the town's own soldiers, armed from its stores). Freed: the town, with its county~~

~~if a seat, returns to its old crown, the surviving garrison is driven home, and both kingdoms chronicle it. Crushed: the rebels die and resentment goes on. After a year with no successful rising the town is integrated~~

- [ ] Captured capitals: let a war or siege take a kingdom's capital (the current rules never allow it), and chronicle the fall from both sides (moved from Step 2)
- [ ] Expansion (Phase 8)
- [ ] Building upgrades: cities spend surplus Wood and Stone to grow buildings' capacity (moved from Step 3, where Wood and Stone went to repairs instead)
- [ ] New kingdoms sprout, rarely: a disloyal lord breaks away with his province (done, through rebellion: lords and provinces part 3), or settlers found a village on empty land (still to do). It starts small, gets a founding monarch and a chronicle that opens with its birth, and grows only through Expansion
- [ ] Nemesis-style grudges: record who killed whom among notable fighters (named champions and commanders); relationships gain a separate grievance value and become directional, so a vendetta can follow one killing (idea from Orpheus)

Step 5: Long-term systems

- [ ] Eras (details under Phase 13 below)
- [ ] Quests that can be accepted and completed (blocks the hero and necromancer eras)
- [ ] Decision Engine (Phase 11)
- [ ] Personality axes for NPCs (aggression, courage, vengefulness, forgiveness, ambition and so on), inherited from parents; added when the Decision Engine starts (idea from Orpheus)
- [ ] Imperfect knowledge: news travels along the roads with delay and fading detail, each kingdom keeps its own account, and a claim records how someone was recognized (personal tie, fame, mistaken match, false testimony); first slice at kingdom level (idea from Orpheus)
- [ ] AI agents playing a few NPCs (Phase 12, plan under the later phases below)

On your side

Three of five setup tasks are done; confirming the push reached GitHub is the one that needs you, and running the check is an ongoing habit.

- [x] ~~Apply the fixed project zip (Weapons typo and Kingdom Favor) - confirmed by check_project.py passing all 12 features~~

- [x] ~~Add check_project.py to the World folder~~
- [x] ~~Add .gitignore and .gitattributes to the repo root, then run the one-time cleanup~~
- [x] ~~Push to GitHub and confirm your repo page shows the latest commit, with no world.db or **pycache** in the file list~~
- [x] ~~Run python World\check_project.py before every commit (ongoing habit)~~

Phases 1-6: the finished core

Everything in these six phases is built and tested; three small items remain open, and the consumption sinks moved into the build order (Step 3).

Phases 1-2: Foundation and simulation core

- [x] ~~SQLite world, schema and migrations~~
- [x] ~~Clock with hour, day, month, season and year ticks~~
- [x] ~~Permanent event log plus Logs/simulation.log~~
- [x] ~~main.py, run_world.py and watch_events.py~~
- [ ] Seed Manager (reproducible worlds): works informally. Small and can be done any time: give each decision its own dice, seeded by the world seed plus the people involved, so results do not depend on processing order. Start with new systems, ideally before armies and campaigns in Step 3 (idea from Orpheus)

Phase 3: World structure

- [x] ~~Roads with real travel times~~
- [x] ~~Kingdom treasury and tax rate~~
- [x] ~~City wealth~~
- [x] ~~Districts~~
- [x] ~~Military upkeep: gold, food and weapons, with desertion if unpaid~~
- [x] ~~Peace treaties after wars~~

Phase 4: NPC life

- [x] ~~Age, Health, Hunger, Energy, Happiness, Stress~~
- [x] ~~Eat, Sleep, Work, Socialize, Heal~~
- [x] ~~Daily schedules: home, work, socializing, eating~~
- [x] ~~Starvation and old-age death~~
- [x] ~~Tourism with happiness-weighted destinations and real treasury income~~

- [x] ~~Migration: desperate families relocate, with pooled gold and a housing check; a ruling household never emigrates~~
- [x] ~~Civil War: rare, needs high stress, low happiness and a weapons surplus~~
- [x] ~~Inter-kingdom travel on the road network~~
- [ ] Walking: NPCs still move between buildings instantly
- [ ] Traveling NPCs still earn wages and healing (deliberate simplification)

Phase 5: Economy

- [x] ~~Stone and Quarry~~
- [x] ~~Food chain: Farm, Grain, Mill, Flour, Bakery, Food~~
- [x] ~~Steel chain: Mine, Ore, Smelter, Steel, Blacksmith (Tools and Weapons)~~
- [x] ~~Cross-kingdom trade with real gold~~
- [x] ~~Caravans that travel real roads and can be raided~~
- [x] ~~Kingdom bias, so each kingdom has an economic character~~

Phase 6: Relationships and memory

- [x] ~~Friend, Rival and Acquaintance graph~~
- [x] ~~Marriages as real relationships~~
- [x] ~~Every relationship remembers its origin~~
- [x] ~~Bonds evolve from shared combat~~
- [x] ~~New bonds form from shared battles~~
- [x] ~~Grief, and widows can remarry (fixed: a stale save was erasing the grief of about half of all widows and widowers, leaving them married to the dead)~~

Phases 7-8: events and politics

Alliances now form, end, and shape which wars get fought; the favor behind them comes from trade, peaceful years, new rulers and wars. Expansion is scheduled in the build order above.

Phase 7: Events and history

- [x] ~~Wedding, Coronation, Raid, War, Death (by cause), Caravan Raided, Civil War~~
- [x] ~~Participants tagged on every combat event~~
- [ ] Drought, Plague, Fire and Flood
- [ ] Festival

- [ ] Crime
- [ ] Monster raids and dragon attacks

Phase 8: Politics and diplomacy

- [x] ~~Revolts (as Civil War)~~
- [x] ~~Peace treaties~~
- [x] ~~Kingdom Favor: directional; each trade gives the buyer +0.3 and the seller -0.1~~
- [x] ~~Alliances: form when both kingdoms reach 60, end when either falls below 40, at most 2 allies each, blocked during a peace treaty~~
- [x] ~~Peace dividend: each peaceful year warms every pair by 3 to 10 in each direction, slowing as favor rises, paused during a peace treaty~~
- [x] ~~Succession shocks, war favor, and grievance-driven wars (see the build order)~~
- [x] ~~Rulers are saved: fixed a bug that crowned a new king or mayor in every city every month~~
- [ ] Crime as a system
- [ ] Prosperity

Phase 13: World history

Each kingdom writes its own chronicle in olde English, dated by year and season, recording big events only. Step 2 of the build order built it: alliances, coronations, ruler deaths, wars from each side and civil wars are all recorded.

Alliance formation

- [x] ~~Yearly check: both kingdoms' favor at or above the threshold (60 for now, tunable)~~
- [x] ~~Set the kingdom relation to Allied~~

Eras (left blank for now, like seasons)

- [ ] Era framework: time-based, event-triggered and quest-triggered
- [ ] At least six eras, each with effects on the world
- [ ] Kingdom-specific notable periods under the world-level era

Later phases and the AI-agent plan

Map, mod support and polish are unscheduled. Armor and sieges (Phase 9), the Decision Engine (Phase 11) and AI agents (Phase 12) are in the build order; the AI-agent plan stays here in detail.

- [ ] Phase 10: Map (paused; caravans, travelers, borders and alliances now exist to draw)
- [ ] Phase 14: Mod support
- [ ] Phase 15: Polish

Phase 12 idea: AI agents playing a few NPCs (idea stage, after the Decision Engine)

- [ ] Flag a few NPCs as agents; everyone else stays rule-driven
- [ ] A shared action menu: the simulation lists legal choices, the agent picks one, the simulation validates and applies it
- [ ] Agent memory stored in the database, built from existing events and relationship origins
- [ ] Agent-to-agent conversations that change relationships
- [ ] Skill teaching between NPCs through the existing skill levels
- [ ] Gossip, so information about events spreads person to person (builds on imperfect knowledge in Step 5)
- [ ] A decision log, so agent runs can be reviewed and replayed

Tooling and idea backlog

The pre-commit check is done and working; the rest here is unscheduled.

Tooling

- [x] ~~check_project.py: config, syntax, features, git hygiene and a simulation smoke test~~
- [x] ~~The git check asks git directly and explains failures (.txt names, UTF-16, missing files)~~
- [ ] World report command: kingdom summaries, favor matrix, alliances
- [ ] Optional git pre-commit hook that runs the check automatically

Idea backlog

- [ ] Research: Scholars in Libraries unlock bonuses
- [ ] Thieves Dens and Smugglers that actually do something
- [ ] Succession crises and heirs

Decisions made so far

These are settled design choices, recorded so we do not relitigate them.

- **Chronicle scope:** big events only. Weddings, trades and ordinary deaths are not recorded; only kings and queens count for deaths.
- **One chronicle per kingdom,** written from that kingdom's point of view in olde-English templates, dated by year and season. Eras stay blank until we build them.
- **Chronicle wording:** templates live in a JSON file you can edit, with three variants per event. The variant is chosen from who and when, not at random, so a replayed world reads the same. Every entry links back to the raw event it tells. Every crowning is recorded (founding monarchs, successors, and those raised after an overthrow), and only a reigning monarch's death is. Wars and civil wars are written before anyone who fell in them.
- **Kingdom Favor is directional.** The gain is three times the loss (+0.3 and -0.1 per trade), so reciprocal trade nets positive for both kingdoms while one-sided giving wears thin, slowly.
- **Favor has four sources:** trade; a peace dividend of 3 to 10 each way per peaceful year that slows as favor rises (nothing at 100) and pauses for 3 years after a war; new rulers; and wars.
- **Alliances:** form when both kingdoms' favor reaches 60, end when either falls below 40, at most two allies per kingdom, none while a peace treaty is active. Checked yearly before wars, so new allies cannot be attacked that year.
- **Succession shocks:** a new monarch keeps half the kingdom's old feelings toward each neighbor and re-rolls the rest (between -40 and 60); the successor of an overthrown ruler keeps a quarter. The very first crowning is not a succession.
- **War by grievance:** a kingdom's yearly chance of war runs from 4% (no enemies) to 12% (a bitter grievance); targets it resents are up to 4 times as likely, ones it likes a quarter as likely. A war costs the defender 30 favor (50 if a city falls) and the attacker 10.
- **Build order:** peacetime politics first, then the chronicle, then the foundations war needs, then real war and Expansion, then the long-term systems. Expansion waits because it needs armies that can move.
- **Armies move as one record** (roster, carried supplies, position on a road), not as hundreds of individual travelers, and are resupplied by caravans.
- **Migration is for the desperate** (high stress). Opportunity-seeking migration waits for the Decision Engine.

- **Cross-kingdom trade only fires on real shortages,** so every trade already counts as aid.
- **Git:** world.db is never versioned, so each device generates its own world. Run check_project.py before every commit.
- **AI agents (idea):** the simulation stays the source of truth and agents only choose from a validated action menu.
- **Orpheus Engine is a design reference only.** A friend's Unreal plugin; our world stays Python and SQLite. We borrow four ideas: imperfect knowledge, personality axes, order-independent decision seeds, and grievance as its own relationship value.